

Positive bidiagonal factorization for every finite-band oscillatory Toeplitz–Hessenberg matrix

Candidate full solution to the question in arXiv:2210.10728

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Status. Candidate full solution, pending expert review.

Source. A. Branquinho, A. Foulquié-Moreno, and M. Mañas, *Positive bidiagonal factorization of tetradiagonal Hessenberg matrices* [1], question on printed page 21.

1 The question

For a lower Hessenberg matrix with r subdiagonals, a *positive bidiagonal factorization* (PBF) has the ordered form

$$T = L_1 L_2 \cdots L_r U, \quad (1)$$

where each L_j is unit lower bidiagonal with strictly positive subdiagonal, while U is upper bidiagonal with unit superdiagonal and strictly positive diagonal. The source proves the Toeplitz case for $r = 2$ and asks in its conclusion:

POSITIVE BIDIAGONAL FACTORIZATION OF TETRADIAGONAL MATRICES

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In [6] for the tetradiagonal case, and corresponding multiple orthogonal polynomials in the step-line with two weights, the PBF factorization is given in terms of the values of the orthogonal polynomials of type I and II at 0 and, consequently, an spectral interpretation of the Darboux transformation is given.

For the future, we will like to understand what happens with Hessenberg matrices with more diagonals. Preliminary attempts show that the role of continued fractions must be replaced by more general objects, maybe branched continued fractions. Similar questions open for the constant Toeplitz matrix, do a oscillatory pentadiagonal Toeplitz matrix always admit a PBF? What happens when the banded recursion matrix has several superdiagonals as well as subdiagonals?

REFERENCES

- [1] Carlos Álvarez-Fernández, Ulises Fidalgo, and Manuel Mañas, *Multiple orthogonal polynomials of mixed type: Gauss–Borel factorization and the multi-component 2D Toda hierarchy*, *Advances in Mathematics* **227** (2011) 1451–1525.
- [2] Alexander Aptekarev, Valery Kaliaguine, and Jeannette Van Iseghem, *The Genetic Sums’ Representation for the Moments of a System of Stieltjes Functions and its Application*, *Constructive Approximation* **16** (2000) 487–524.
- [3] Amílcar Branquinho, Juan E. Fernández-Díaz, Ana Foulquié-Moreno, and Manuel Mañas, *Hypergeometric Multiple*

Figure 1: The exact question in the original paper, printed page 21. This is a full-width crop rendered from the archived source PDF.

Transcription. “Similar questions open for the constant Toeplitz matrix, do a oscillatory pentadiagonal Toeplitz matrix always admit a PBF?”

The answer is yes. In fact, the number of subdiagonals is arbitrary.

2 Proof intuition

The Edrei–Schoenberg classification turns total nonnegativity of a banded Toeplitz matrix into a factorization of a polynomial:

$$1 + e_1 z + \cdots + e_{r+1} z^{r+1} = \prod_{j=1}^{r+1} (1 + \beta_j z), \quad \beta_j > 0.$$

Writing S for the unilateral lower shift immediately factors the matrix as one upper factor followed by r lower factors. This is exactly the wrong order for (1).

The upper factor can nevertheless be moved rightward, one lower factor at a time, without losing positivity. After the first m roots have been absorbed, its n th diagonal entry is

$$q_n^{(m)} = \frac{h_{n+1}(\beta_1, \dots, \beta_m)}{h_n(\beta_1, \dots, \beta_m)},$$

where h_n is the complete homogeneous symmetric polynomial. The elementary recurrence for h_n is precisely the quotient–difference identity needed to exchange an upper and a lower bidiagonal factor. All h_n are positive, so every new factor is positive. Iterating gives the requested order.

3 Statement and proof

Index rows and columns by $\mathbb{N}_0$. Let S be the unilateral lower shift, $S_{n,n-1} = 1$ for $n \geq 1$; thus $S^* S = I$. For $r \geq 1$ consider

$$T = S^* + e_1 I + e_2 S + \cdots + e_{r+1} S^r. \quad (2)$$

This is a Toeplitz–Hessenberg matrix with $r + 2$ nonzero diagonals.

Theorem 1. *For every $r \geq 1$, an oscillatory Toeplitz–Hessenberg matrix (2) admits a PBF (1). More explicitly, if*

$$1 + e_1 z + \cdots + e_{r+1} z^{r+1} = \prod_{j=1}^{r+1} (1 + \beta_j z), \quad \beta_j > 0, \quad (3)$$

then the entries of all factors are given by positive quotients of complete homogeneous polynomials, as in (4) below.

Proof. The Edrei–Schoenberg theorem [2, 3], applied exactly as in the source’s tetradiagonal Toeplitz argument, says that oscillation of (2) yields (3); positivity of the outermost subdiagonal makes every β_j strictly positive.

For $m \in \{1, \dots, r + 1\}$ and $n \geq 0$, set

$$H_n^{(m)} := h_n(\beta_1, \dots, \beta_m), \quad q_n^{(m)} := \frac{H_{n+1}^{(m)}}{H_n^{(m)}}.$$

Here $H_0^{(m)} = 1$, and every $H_n^{(m)} > 0$. Let $U^{(m)}$ be the upper bidiagonal matrix with diagonal $(q_n^{(m)})_{n \geq 0}$ and unit superdiagonal. In particular, $U^{(1)} = S^* + \beta_1 I$. Write $L(\beta) = I + \beta S$.

For $m \geq 2$, define a unit lower bidiagonal matrix B_{m-1} by

$$(B_{m-1})_{n,n-1} = \lambda_n^{(m)} := \frac{\beta_m q_n^{(m-1)}}{q_{n-1}^{(m)}} > 0, \quad n \geq 1. \quad (4)$$

We claim the local exchange identity

$$U^{(m-1)}L(\beta_m) = B_{m-1}U^{(m)}. \quad (5)$$

Both sides have unit superdiagonal and no entries below the first subdiagonal. Their $(n, n-1)$ entries agree by (4). At $n=0$, the diagonal equality is $q_0^{(m)} = q_0^{(m-1)} + \beta_m$. For $n \geq 1$, use

$$H_n^{(m)} - \beta_m H_{n-1}^{(m)} = H_n^{(m-1)}, \quad H_{n+1}^{(m)} = H_{n+1}^{(m-1)} + \beta_m H_n^{(m)}. \quad (6)$$

Indeed,

$$\begin{aligned} q_n^{(m-1)} + \beta_m - \lambda_n^{(m)} &= \frac{H_{n+1}^{(m-1)}}{H_n^{(m-1)}} \left(1 - \frac{\beta_m H_{n-1}^{(m)}}{H_n^{(m)}} \right) + \beta_m \\ &= \frac{H_{n+1}^{(m-1)}}{H_n^{(m)}} + \beta_m = \frac{H_{n+1}^{(m)}}{H_n^{(m)}} = q_n^{(m)}. \end{aligned}$$

This proves (5), including strict positivity of its new lower factor.

Finally, $S^*S^k = S^{k-1}$ for $k \geq 1$, so expansion of (3) gives the exact initial factorization

$$T = U^{(1)}L(\beta_2)L(\beta_3) \cdots L(\beta_{r+1}). \quad (7)$$

Apply (5) successively for $m = 2, 3, \dots, r+1$:

$$T = B_1U^{(2)}L(\beta_3) \cdots L(\beta_{r+1}) = \cdots = B_1B_2 \cdots B_rU^{(r+1)}.$$

Every subdiagonal entry of every B_j and every diagonal entry of $U^{(r+1)}$ is strictly positive. This is the required PBF. $\square$

Corollary 1 (answer to the source question). *Every oscillatory pentadiagonal Toeplitz–Hessenberg matrix admits a positive bidiagonal factorization with three unit lower bidiagonal factors followed by one positive upper bidiagonal factor.*

Proof. Take $r = 3$ in Theorem 1. $\square$

4 Verification, scope, and novelty

The exact-arithmetic script `code/verify_pbf.py` constructs the factors from (4) and compares their product entrywise with twelve-by-twelve leading Toeplitz sections. It checks five cases from one through five subdiagonals, including repeated roots. All checks pass. This is a normalization check only; equations (6)–(7) constitute the proof.

The result concerns the source’s one-superdiagonal Toeplitz–Hessenberg setting. It does not answer the source’s broader final question about matrices having several superdiagonals as well as several subdiagonals, nor does it claim that arbitrary non-Toeplitz oscillatory Hessenberg matrices have a PBF.

Novelty check. On 12 August 2026, bounded arXiv and run-index searches used the exact title and source question together with “pentadiagonal Toeplitz PBF”, “banded Toeplitz bidiagonal factorization”, “quotient–difference”, “complete homogeneous”, and “branched continued fractions”. Later arXiv:2412.03694 and arXiv:2603.21345 develop broad MOP/branched factorizations, but source inspection and exact searches found neither the Toeplitz theorem nor formula (4). Novelty confidence is moderate rather than definitive.

Human-review recommendation. Prioritize review. The main checks are the orientation of the source’s PBF convention and the general finite-polynomial Edrei–Schoenberg invocation; the quotient–difference calculation itself is elementary and exact.

References

- [1] A. Branquinho, A. Foulquié-Moreno, and M. Mañas, *Positive bidiagonal factorization of tetradiagonal Hessenberg matrices*, arXiv:2210.10728 (2022).
- [2] A. Edrei, *Proof of a conjecture of Schoenberg on the generating function of a totally positive sequence*, *Canad. J. Math.* **5** (1953), 86–94.
- [3] I. J. Schoenberg, *Some analytical aspects of the problem of smoothing*, in *Studies and Essays Presented to R. Courant on His 60th Anniversary*, Interscience, 1948, 351–370.